

Project 1: CVDs predictions (Team ML212)

Leila Benjelloun

leila.benjelloun@epfl.ch

Rania Hatrouhou

rania.hatrouhou@epfl.ch

Ismael Berrada

ismael.berrada@epfl.ch

1 Introduction

This project addresses the challenge of predicting cardiovascular diseases (CVDs) from large-scale health survey data. The dataset comprises 328,135 samples with 321 features from telephone health surveys, presenting a severely imbalanced binary classification problem with only 8.83% positive CVD cases. This imbalance, combined with extensive missing data and heterogeneous feature types, requires careful methodological choices to achieve meaningful predictive performance.

2 Methodology

2.1 Data Understanding and Preprocessing Strategy

Our exploratory analysis revealed a dataset with 91.17% negative cases, extensive missingness (239 features affected, up to 99.99% missing), and mixed data types: 101 categorical survey responses, 73 continuous measurements, and 70 binary features. We identified survey-specific missing codes (7, 9, 77, 99) and removed 8 non-predictive date/ID features.

Rather than applying a single preprocessing approach, we designed a systematic pipeline with increasing sophistication levels to understand the trade-offs between data retention and feature quality. This approach was motivated by the hypothesis that different preprocessing strategies would significantly impact model performance on this challenging dataset.

2.2 Experimental Design: A Systematic Three-Phase Approach

Given the strong class imbalance in our dataset, we use the F1-score (see ??), as our primary evaluation metric, as it provides a balanced measure of precision and recall and more accurately reflects model performance on the minority class.

Phase 1: Preprocessing Impact Assessment We fixed logistic regression as our baseline model and systematically evaluated preprocessing levels 0-4 using 5-fold stratified cross-validation [check detailed levels 1].

Phase 2: Model Selection with Fixed Preprocessing

After identifying the best preprocessing level, we compared logistic regression, regularized logistic regression, and random forest using comprehensive grid search. This phase tested our hypothesis that different model architectures would perform differently on the preprocessed data.

Phase 3: Alternative Imbalance Handling Recognizing that Level 2 uses oversampling, we explored cost-sensitive learning with Level 1 preprocessing to compare different approaches to class imbalance. This phase addressed the question of whether algorithmic solutions (class weights) could match data-level solutions (oversampling).

Phase 4: Fine-Tuning In the final phase, we focused on fine-tuning the best-performing model identified during our comparison stage. This step involved refining key hyper-parameters to maximize predictive performance while preventing overfitting.

3 Results and Analysis

3.1 Preprocessing Impact: From Failure to Success

As shown in Figure 1, performance improved substantially with increasingly rigorous preprocessing. In particular, Level 2 achieved the best overall balance across all metrics (F1 = 0.638, ROC-AUC = 0.838, PR-AUC = 0.712, Precision = 0.696, Recall = 0.588), highlighting the importance of moderate feature cleaning and encoding. Beyond this level, performance declined slightly, suggesting that overly aggressive preprocessing (e.g., high feature dropping or heavy smoothing) may remove informative variability crucial for classification.

We therefore adopt Level 2 as our standardized preprocessing configuration for the subsequent phases of the study.

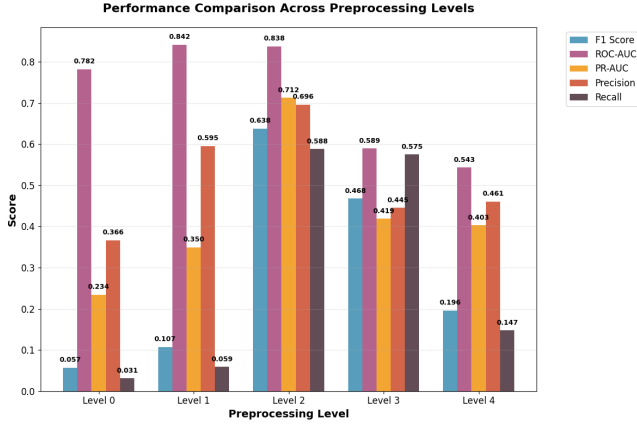


Figure 1: Performance comparison across preprocessing levels (L0–L4) showing F1-Score, ROC-AUC, PR-AUC, Precision, and Recall metrics.

3.2 Model Architecture Comparison

With Level 2 preprocessing fixed, our model comparison revealed clear performance hierarchies:

Regularized logistic regression emerged as the optimal choice (F1=0.649), outperforming both standard logistic regression (F1=0.638) and random forest (F1=0.546). The L2 regularization ($\lambda = 0.001$) provided the optimal bias-variance trade-off, preventing overfitting while maintaining predictive power.

Interestingly, while random forest achieved the highest precision (0.723), it suffered from lower recall, resulting in inferior F1-scores. This suggests that ensemble methods, while theoretically powerful, may not be optimal for this specific combination of data characteristics and preprocessing approach.

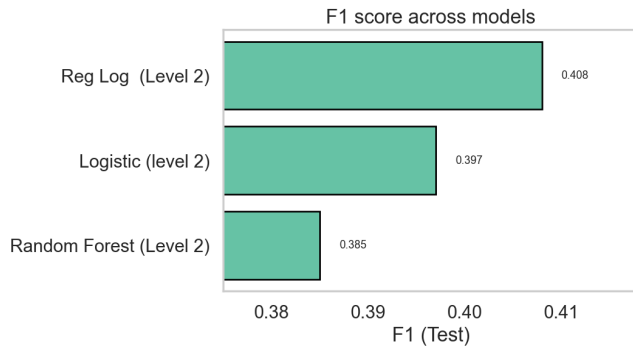


Figure 2: Performance comparison across test sets of different models showing obtained F1 scores on AICrowd.

3.3 Class Imbalance Handling: Oversampling vs. Cost-Sensitive Learning

Figure 3 compares two strategies for addressing class imbalance: oversampling (Level 2) and cost-sensitive learning using Level 1. The regularized logistic model trained on Level 2 preprocessing achieved the highest F1

score (0.408), outperforming both weighted logistic and weighted regularized logistic approaches. This suggests that effective preprocessing and balanced feature representation had a stronger impact on handling imbalance than weighting schemes alone.

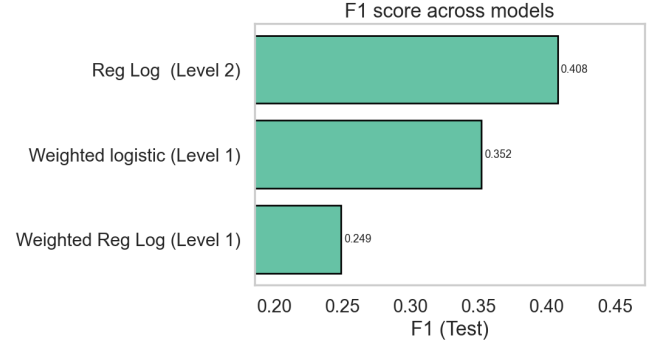


Figure 3: Performance comparison across test sets of different weighted models showing obtained F1 scores on AICrowd.

3.4 Final Model Optimization

Given that regularized logistic regression with Level 2 preprocessing demonstrated the best performance, we proceeded to conduct an extensive hyperparameter optimization to further refine the model.

Grid Search Configuration: The grid search was performed using 5-fold cross-validation on the training set, exploring the following hyperparameter ranges:

Hyperparameter	Values Tested
Learning rate (γ)	{0.05, 0.08, 0.10, 0.15}
Regularization strength (λ)	{0.0003, 0.0005, 0.001, 0.002, 0.005}
Maximum iterations (max_iters)	3000

This grid search configuration yielded our best-performing model, achieving an **F1-score of 0.412** and an **accuracy of 0.845**, confirming the effectiveness of fine-tuning regularized logistic regression under Level 2 preprocessing.

4 Conclusion

Our approach yielded several key insights: (1) preprocessing quality emerged as the primary determinant of model success, (2) L2 regularization provided the best bias–variance balance for linear models, and (3) over-sampling proved more effective than cost-sensitive learning for handling imbalance. Overall, these findings suggest that in imbalanced medical datasets with extensive missing data, the sophistication of data preprocessing may play a more critical role than model complexity itself.

A Appendix

A.1 Preprocessing Levels Detailed

Table 1 summarizes the parameters used for each preprocessing configuration (L0–L4). Each level represents an increasingly aggressive cleaning and encoding strategy, progressively tightening missing-value thresholds, low-variance removal, and target encoding strength.

Table 1: Preprocessing configurations across levels (L0–L4).

Level	Missing $\leq$	Drop low var	Lowvar th.	Cont. impute	Cat. impute	Drop dups
L0	0.99	✗	–	mean	mode	✗
L1	0.80	✓	0.999	mean	mode	✗
L2	0.60	✓	0.995	mean	mode	✗
L3	0.40	✓	0.990	median	mode	✓
L4	0.60	✓	0.995	mean	mode	✓

Level	TE min	TE max cats	TE smooth
L0	0	0	1.0
L1	100	50	1.0
L2	50	20	1.0
L3	20	10	0.5
L4	50	20	0.5

A.2 Evaluation Metrics: F1-Score, PR-AUC, and ROC-AUC

F1-Score. The F1-score is the harmonic mean of precision and recall, defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

It provides a balanced measure of a model’s ability to correctly identify positive cases while limiting false positives. In imbalanced datasets, where accuracy can be misleading due to the dominance of the majority class, the F1-score offers a more informative evaluation by emphasizing performance on the minority (positive) class.

PR-AUC (Precision–Recall Area Under Curve). The PR-AUC represents the area under the Precision–Recall curve, which plots precision against recall at varying classification thresholds. It summarizes how well a model maintains high precision and recall simultaneously. Unlike ROC-AUC, PR-AUC is particularly sensitive to class imbalance, making it a valuable indicator when the positive class is rare.

ROC-AUC (Receiver Operating Characteristic Area Under Curve). The ROC-AUC measures the area under the ROC curve, which plots the True Positive Rate (Recall) against the False Positive Rate across thresholds. It evaluates a model’s overall ability to discriminate between classes. A higher ROC-AUC indicates

stronger separability between positive and negative samples, though its interpretation can be less reliable when class imbal

A.3 Model Performance Comparison Table

Table 2 presents a comprehensive comparison of all tested models across different preprocessing levels, showing key performance metrics obtained through 5-fold cross-validation.

Table 2: Model Performance Comparison Across Preprocessing Levels During Training

Model	Preproc.	F1	PR-AUC	ROC-AUC	Prec.
Raw Data Results					
Log. Regression	Raw	0.000	0.089	0.503	0.000
Reg. Logistic	Raw	0.000	0.089	0.503	0.000
Level 2 Preprocessing Results					
Log. Regression	L2	0.638	0.712	0.838	0.696
Reg. Logistic	L2	0.649	0.727	0.846	0.706
Random Forest	L2	0.546	0.691	0.824	0.723
Level 1 + Weighted Results					
Weighted Reg. Log.	L1	0.245	0.358	0.845	0.505

A.4 Precision–Recall Trade-off Analysis

Figure 4 illustrates the precision–recall trade-off across different preprocessing levels, highlighting how Level 2 achieves the best balance between precision and recall for this imbalanced dataset.

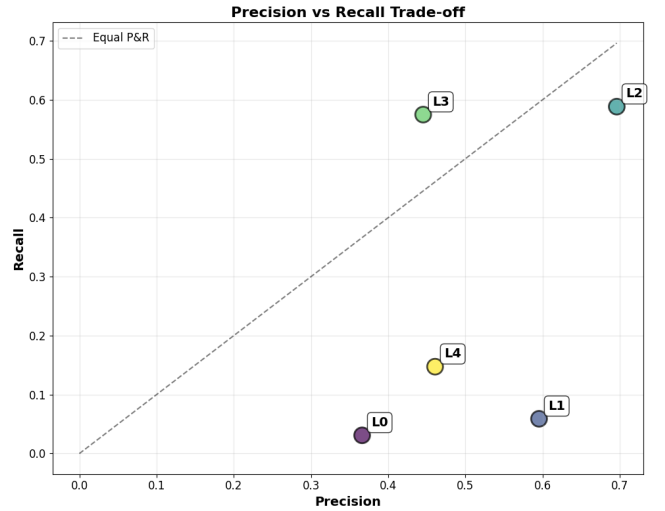


Figure 4: Precision vs. Recall trade-off across preprocessing levels (L0–L4).